BENETECH SOLUTIONS

Enterprise Infrastructure Planning for a Startup Company

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ITEP 414 - System Administration and Maintenance
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Bachelor of Science in Information Technology

PART 1 - COMPANY PROFILE

Company Name: BeneTech Solutions

Nature of Business

BeneTech Solutions is a startup software development company that provides digital solutions for businesses and organizations. The company focuses on developing web applications, business management systems, and customized software based on client needs.

Office Location

Santa Cruz, Laguna, Philippines

BeneTech Solutions is located in Santa Cruz, Laguna. The location provides the company with a suitable environment for its software development operations while allowing it to serve businesses and organizations within Laguna and nearby areas.

Vision

To become a trusted software development company that provides practical, reliable, and innovative digital solutions that help businesses and organizations improve their daily operations.

Mission

BeneTech Solutions aims to develop user-friendly and reliable software solutions based on the needs of its clients. The company is committed to providing quality services, continuously improving its technical skills, and using modern technologies to help organizations work more efficiently.

Organizational Structure

BeneTech Solutions has a simple organizational structure suitable for a startup company with 20 employees. The employees are divided into four main departments: Information Technology, Human Resources, Finance, and Sales.

Department	Positions	Employees
Information Technology	1 IT Manager / Senior Developer; 2 Software Developers; 1 Junior System Administrator; 1 Helpdesk / Technical Support Technician	5
Human Resources	1 HR Manager; 1 Recruitment Officer; 1 HR Coordinator; 1 Administrative Assistant	4
Finance	1 Finance Manager; 2 Accountants; 1 Payroll Officer; 1 Finance Assistant	5
Sales	1 Sales Manager; 3 Sales Representatives; 1 Account Executive; 1 Sales Coordinator	6
Total		**20**

PART 2 - ENTERPRISE HARDWARE INVENTORY

BeneTech Solutions requires reliable computer hardware and supporting infrastructure for its 20 employees. The hardware was selected based on the requirements of each department, expected workload, equipment reliability, data protection, and future expansion.

Asset ID	Hardware	Qty	Department / Area	Purpose
HW-001	Desktop Computers	13	IT, HR, Finance, Sales	Primary workstations for development, accounting, administrative work, and daily office operations.
HW-002	Laptops	7	IT, HR, Finance, Sales	Portable computers for managers and employees who require mobility, meetings, presentations, or client interaction.
HW-003	Server	1	IT / Server Area	Centralized services and resources for the company network.
HW-004	Router	1	IT / Network Area	Connects the internal network to the internet and routes traffic.
HW-005	Managed Gigabit Switch	1	IT / Network Area	Provides wired connectivity for computers, server, printer, access points, and other devices.
HW-006	Network Printer	2	Shared Office Areas	Shared document printing without separate printers per department.
HW-007	UPS	3	Server & Network Area	Temporary backup power and protection for critical infrastructure.
HW-008	Wireless Access Point	2	Office Area	Wi-Fi connectivity throughout the single office floor.
HW-009	NAS Storage	1	IT / Server Area	Centralized shared storage and internal backups.
HW-010	External Backup Drive	2	IT Department	Additional backup copies of important company data.
HW-011	Monitors	20	All Departments	13 primary desktop displays plus 7 additional displays for selected staff.

Recommended Hardware Specifications

Desktop Computers: Intel Core i5 or AMD Ryzen 5; 16 GB RAM; 512 GB SSD; Gigabit Ethernet; Windows 11 Pro support.

Laptops: Intel Core i5 or AMD Ryzen 5; 16 GB RAM; 512 GB SSD; 14-15.6 inch Full HD; Wi-Fi 6.

Server: Intel Xeon E-series or equivalent; 32 GB ECC RAM; 2 x 2 TB enterprise drives; RAID 1; dual Gigabit Ethernet; Ubuntu Server.

Managed Switch: 24 Gigabit Ethernet ports; managed capability; VLAN; QoS; link aggregation; web management.

Router: Gigabit WAN/LAN; DHCP; NAT; VPN; traffic monitoring; access control.

NAS: 4-bay enclosure; 4 x 4 TB NAS-grade drives; RAID support; Gigabit Ethernet; access controls; scheduled backup.

Monitors: 20 units; approximately 24-inch Full HD; HDMI/DisplayPort where possible.

PART 3 - ENTERPRISE SOFTWARE INVENTORY

BeneTech Solutions requires operating systems, productivity applications, development tools, security software, and system administration utilities to support daily operations.

Software	Version	License	Purpose
Windows 11 Pro	64-bit, Current Release	Commercial	Primary OS for employee desktops and laptops.
Ubuntu Server	LTS	Open Source	Server OS for internal company services.
Microsoft 365	Current Release	Commercial Subscription	Word, Excel, PowerPoint, Outlook, and productivity tools.
Visual Studio Code	Current Release	Free	Code editor for development and configuration files.
Git	Current Release	Open Source	Tracks changes in source code and projects.
GitHub Desktop	Current Release	Free	Graphical interface for Git repositories and workflows.
Oracle VirtualBox	Current Release	Open Source / Free	Virtual machines for testing systems and configurations.
Google Chrome	Current Release	Freeware	Web applications, research, and website testing.
Microsoft Defender	Included with Windows 11	Included with Windows	Antivirus and malware protection.
AnyDesk	Current Release	Commercial / Business License	Authorized remote technical support.
7-Zip	Current Release	Open Source	Compresses and extracts files and archives.

These selections balance compatibility, productivity, development needs, security, and cost. Windows 11 Pro supports business workstations, Ubuntu Server provides a reliable server platform, Microsoft 365 supports office productivity, and the remaining tools support development, virtualization, browsing, security, remote support, and file management.

PART 4 - ENTERPRISE NETWORK INVENTORY

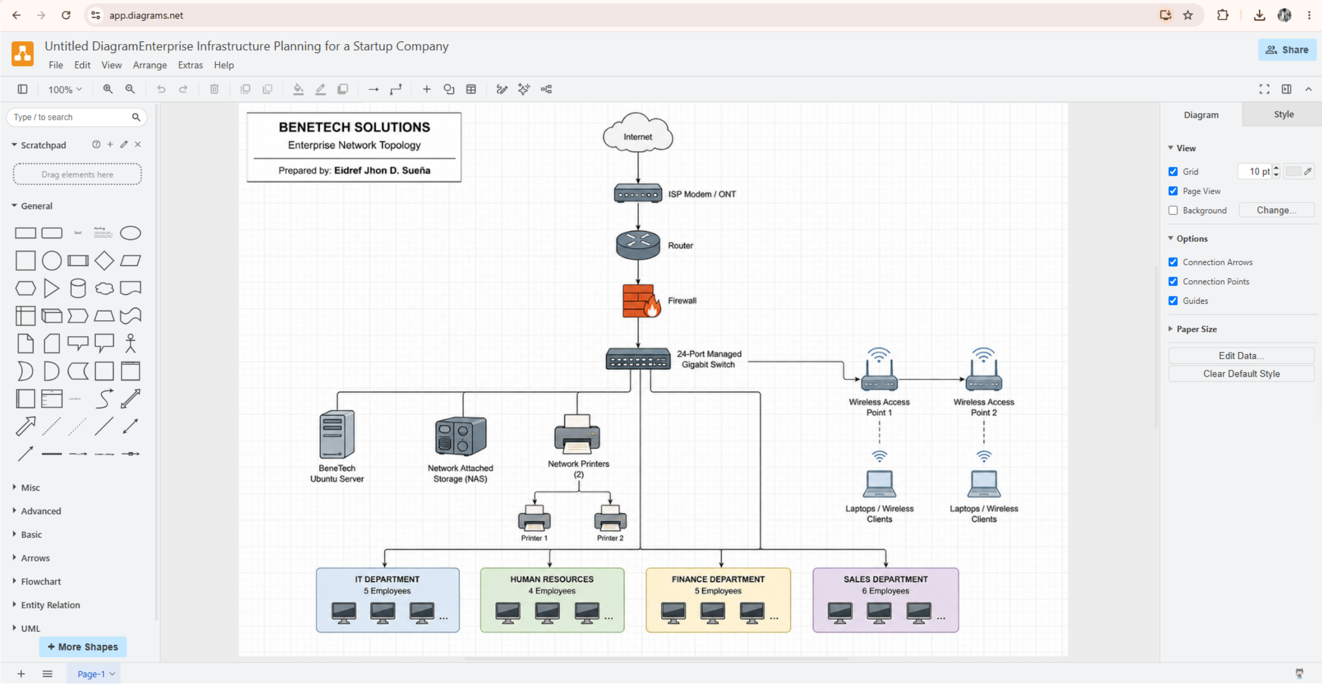
The proposed network infrastructure connects BeneTech Solutions' 20 employees, server, shared storage, printers, and wireless devices across a single office floor.

Asset ID	Equipment	Qty	Specification	Purpose
NET-001	ISP Modem / ONT	1	Gigabit-capable ISP unit	Receives the ISP connection.
NET-002	Router	1	Gigabit WAN/LAN, DHCP, NAT, VPN	Routes internal and external traffic.
NET-003	Firewall	1	Gigabit firewall appliance	Filters traffic according to security rules.
NET-004	Managed Network Switch	1	24-Port Gigabit Managed Switch	Provides wired network connectivity.
NET-005	Wireless Access Point	2	Wi-Fi 6, dual-band, WPA2/WPA3	Provides wireless coverage.
NET-006	Patch Panel	1	24-Port CAT6 Patch Panel	Organizes Ethernet terminations.
NET-007	CAT6 Ethernet Cable	1 Box	305-meter CAT6 box	Wired Ethernet connections.
NET-008	RJ45 Connectors	50	CAT6-compatible	Cable termination and maintenance.

The network follows a simple and logical structure: Internet -> ISP Modem/ONT -> Router -> Firewall -> Managed Switch. The switch then distributes connectivity to the server, NAS, printers, wireless access points, and the four departments.

PART 5 - ENTERPRISE NETWORK DIAGRAM

The following network topology represents the proposed BeneTech Solutions infrastructure and shows the logical connection of the internet, modem/ONT, router, firewall, managed switch, server, NAS, printers, wireless access points, wireless clients, and four departments.



PART 6 - SYSTEM ADMINISTRATION ROLES

1. Helpdesk Technician

Responsibilities: Provides first-line technical support for hardware, software, accounts, printers, and basic network problems.

Skills: Hardware/software troubleshooting, Basic networking, Windows administration, Communication and documentation.

Common Tools: Remote support software, Ticketing systems, PowerShell / Command Prompt, Microsoft Defender.

Certifications: CompTIA A+, CompTIA Network+, Microsoft Fundamentals certifications.

2. Network Administrator

Responsibilities: Manages and maintains network infrastructure to keep devices connected reliably and securely.

Skills: TCP/IP and subnetting, Routing and switching, VLANs, Firewall fundamentals, Wireless networking.

Common Tools: Cisco IOS, Wireshark, PuTTY, Ping / Traceroute, Network monitoring tools.

Certifications: CCNA, CompTIA Network+, CompTIA Security+.

3. Linux System Administrator

Responsibilities: Manages Linux servers and keeps them available, secure, updated, backed up, and properly configured.

Skills: Linux administration, Command line, Permissions, Shell scripting, Server configuration, Troubleshooting.

Common Tools: Linux Terminal, SSH, Bash, systemctl, APT, Cron.

Certifications: CompTIA Linux+, LPIC-1, RHCSA.

4. Cloud Administrator

Responsibilities: Manages cloud-based virtual machines, storage, access, networking, monitoring, and security.

Skills: Cloud fundamentals, Virtualization, Networking, Identity and access management, Cloud security, Automation.

Common Tools: Microsoft Azure, Amazon Web Services, Google Cloud, PowerShell, Linux CLI.

Certifications: Microsoft Azure Administrator Associate, AWS Certified CloudOps Engineer - Associate, Google Associate Cloud Engineer.

How the Four Professionals Work Together

Helpdesk Technicians, Network Administrators, Linux System Administrators, and Cloud Administrators work together to maintain reliable IT operations. Helpdesk staff handle direct user issues and escalate network, server, or cloud problems to the appropriate specialist. Network Administrators maintain connectivity, Linux System Administrators manage server systems, and Cloud Administrators manage cloud resources. By sharing technical information and coordinating responsibilities, the team can resolve problems efficiently and keep company technology available and secure.

PART 7 - INFRASTRUCTURE RECOMMENDATIONS

The following recommendations are based on the needs of BeneTech Solutions as a 20-employee startup. They are intended to provide a reliable, secure, and scalable foundation for current operations and future growth.

Recommendation	Details	Reason / Benefit
Internet Provider	Use a reliable business fiber connection with sufficient bandwidth for development, cloud access, video meetings, and file transfers. A secondary connection from another provider can be considered for continuity.	Reliable connectivity is essential for daily operations; redundancy reduces downtime.
Server Specifications	Use an on-premise mid-range server with a server-grade processor, 32 GB ECC RAM, redundant storage, Gigabit networking, and Ubuntu Server LTS.	ECC memory and redundant storage improve reliability and data integrity while leaving room for growth.
Backup Strategy	Perform daily automated backups to NAS, rotate external backup drives, and keep an additional copy separate from the primary systems. Test restoration regularly.	Multiple backup copies reduce the impact of hardware failure, accidental deletion, malware, or other incidents.
Security Recommendations	Use the firewall, network segmentation where practical, regular patching, least-privilege access, two-factor authentication for critical accounts, and staff security awareness.	Layered controls reduce unauthorized access and limit the impact of security incidents.
Antivirus	Use Microsoft Defender on Windows endpoints and keep protection, signatures, and operating systems updated.	Provides baseline endpoint protection without unnecessary complexity for a small startup.
Password Policy	Require long, unique passwords or passphrases; prevent password sharing and reuse; use account lockout and multi-factor authentication for sensitive accounts.	Strong authentication reduces the risk of unauthorized access.
Expansion Plan	Reserve switch capacity, choose expandable server/NAS hardware, add access points if coverage demands increase, and upgrade internet bandwidth as staff and workloads grow.	Planning ahead allows BeneTech to expand without replacing the entire infrastructure.

PART 8 - PERSONAL REFLECTION

Completing the Enterprise Infrastructure Planning project helped me understand that building an IT infrastructure involves much more than simply purchasing computers and connecting them to the internet. Before starting this activity, I already had some knowledge about computers, networking, operating systems, and different software tools. However, this project helped me understand how these technologies can be planned and organized together to support the operations of an actual company.

One of the most important things I learned was how to identify the hardware, software, and networking requirements of an organization based on its number of employees and their responsibilities. While planning the infrastructure for BeneTech Solutions, I had to consider what equipment each department would need, how employees would connect to the network, what software would support their work, and how company data could be stored and protected. I also learned more about the responsibilities of different IT professionals, including Helpdesk Technicians, Network Administrators, Linux System Administrators, and Cloud Administrators.

The most challenging part of the project for me was designing the enterprise network topology. I had to think carefully about how the internet connection, modem, router, firewall, switch, server, wireless access points, printers, and different departments should be connected. Creating the diagram helped me understand the flow of a network better because I was able to visualize how different devices communicate instead of only reading about them.

I also realized why infrastructure planning is important before actual deployment. Without proper planning, a company may purchase unnecessary equipment, experience network problems, create security risks, or have difficulty expanding its infrastructure in the future. A clear infrastructure plan helps determine what resources are needed and how they should be implemented before money and time are spent.

Overall, this project gave me a better understanding of the responsibilities of a System Administrator. It allowed me to practice infrastructure planning, hardware and software selection, network design, documentation, and security planning. These skills will help me become more prepared to manage and maintain IT systems and make better technical decisions when working in a real organization in the future.

End of Enterprise Infrastructure Plan